

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 6342A	Course Title: Introduction to Artificial Intelligence and Machine Learning
Semester: 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

To impart fundamental understanding of artificial intelligence and machine learning concepts.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Mathematics		Mathematics	1,2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of Artificial Intelligence	13	Understanding
CO2	Apply Mathematical logic for knowledge representation of AI	15	Understanding
CO3	Demonstrate supervised machine learning techniques	15	Understanding
CO4	Demonstrate Unsupervised Learning techniques	15	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Descriptions	Duration (Hours)	Cognitive Level
-----------------	--------------	------------------	-----------------

CO1	Explain the concept of Artificial Intelligence		
M1.01	Define AI, foundation and history of AI.	2	Understanding
M1.02	Discuss applications of AI.	2	Understanding
M1.03	Explain the Turing test.	3	Understanding
M1.04	Explain AI problems	3	Understanding
M1.05	Explain agents and environments.	2	Understanding
Contents : Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents.			
CO2	Apply Mathematical logic for knowledge representation of AI		
M2.01	Summarize Ontologies, foundations of knowledge representation and reasoning	2	Understanding
M2.02	Describe the representation and reasoning about objects, relations, events, actions, time and space	2	Understanding
M2.03	Demonstrate Propositional Logic	3	Understanding
M2.04	Demonstrate First Order Logic	2	Understanding
M2.05	Explain Soundness and Completeness	3	Understanding
M2.06	Explain Forward and Backward chaining	3	Understanding
Contents: Knowledge representation and reasoning-Ontologies,foundations of knowledge representation and reasoning, representing and reasoning about objects,relations,events,actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Soundness and Completeness, Forward and Backward chaining.			
CO3	Demonstrate supervised machine learning techniques		

M3.01	Explain the Machine Learning concepts.	3	Understanding
M3.02	Describe the process involved in machine learning techniques.	3	Understanding
M3.03	Describe Supervised Learning	3	Understanding
M3.04	Demonstrate Classification techniques	3	Understanding
M3.05	Demonstrate Regression techniques	3	Understanding
Contents: Machine Learning - Definition -Applications - Processes involved in Machine Learning, Machine Learning Techniques-Supervised Learning, Unsupervised Learning and Reinforcement Learning, Data preprocessing – Normalization, Missing value handling. Supervised Learning Classification - Learning a Class from Examples, Logistic Regression - K-Nearest Neighbours - Support Vector Machine - Naive Bayes - Decision Tree - Random Forest Regression- Linear Regression -Multi Linear Regression			
CO4	Demonstrate Unsupervised Learning techniques		
M4.01	Describe types of Clustering	3	Understanding
M4.02	Demonstrate Clustering techniques	6	Understanding
M4.03	Illustrate Dimensionality Reduction techniques	6	Understanding
Contents: Clustering - Discovering clusters - Types of Clustering: Hierarchical, Agglomerative Clustering, Divisive clustering, Partitional Clustering - K-means clustering, Mean Shift clustering Dimensionality reduction – Principal Component Analysis, Factor Analysis, Multidimensional scaling, Linear Discriminant Analysis.			

Text / Reference

T/R	Book Title/Author
T1	Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd Edition. Prentice Hall.
T2	Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd.
T3	Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010.
T4	Tom Mitchell, Machine Learning, McGraw-Hill, 1997
R1	Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill
R2	Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India)

Online Resources

Sl.No	Website Link
1	https://onlinecourses.swayam2.ac.in/aic20_sp06/preview
2	Introduction to Machine Learning - Course (nptel.ac.in)